

Cross-Platform Command Line Reference

Bash/zsh vs. PowerShell CLI Environments & Flag Equivalents

The CLI Portability Rule: Terminal environments use different background layout syntax maps depending on the host OS. While Git commands remain identical everywhere, standard navigation shortcuts and modifier flags change. This reference guides you through mapping Unix commands (macOS/Linux/Git Bash) into native Windows PowerShell expressions safely.

1. Directory Navigation & Path Traversals

Operation	Bash & zsh (macOS / Linux)	Windows PowerShell Equivalence
Print Active Folder	Prints the absolute path string of the current working directory node.	<code>pwd</code>
Change Directory	Moves your cursor context into a target folder path location.	<code>cd folder_name</code>
Jump to User Home	Returns path context instantly back to your user profile root directory.	<code>cd ~</code>
Step Up One Level	Moves workspace context one parent directory hierarchy node upward.	<code>cd ..</code>

2. Directory Mapping & Behavioral Modifier Flags

Listing Strategy	Bash & zsh (macOS / Linux)	Windows PowerShell Syntax Maps
Standard Flat View	<code>ls</code>	<code>ls</code> # Maps to <code>Get-ChildItem</code> under the hood
Detailed Long List	<code>ls -l</code>	<code>dir (or) Get-ChildItem</code>
Expose Hidden Assets	<code>ls -a</code>	<code>ls -Force</code>
Combined Long + Hidden	<code>ls -la</code>	<code>Get-ChildItem -Force</code>

3. File Operations, Creation, & Deletion Arrays

FileSystem Modification	Bash & zsh (macOS / Linux)	Windows PowerShell Syntax Maps
Spawn Empty Document	<code>touch analysis.py</code>	<code>New-Item analysis.py</code> # Alternative: <code>echo \$null > analysis.py</code>
Generate Directory	<code>mkdir src</code>	<code>mkdir src</code>

FileSystem Modification	Bash & zsh (macOS / Linux)	Windows PowerShell Syntax Maps
Delete Isolated File	<code>rm analysis.py</code>	<code>rm analysis.py</code> # Native alias for Remove-Item
Purge Folder Recursively	<code>rm -rf src</code>	<code>rm -Recurse -Force src</code>

4. Environmental Verification Smoke Tests

These commands execute identically across both operating system layers, as they tap into your language environment pointers directly.

Inspection Focus	Bash & zsh (macOS / Linux)	Windows PowerShell Syntax Maps
Verify Python Runtime	<code>python --version</code> # Some Unix environments require: <code>python3 --version</code>	<code>python --version</code>
Verify Git Engine	<code>git --version</code>	<code>git --version</code>